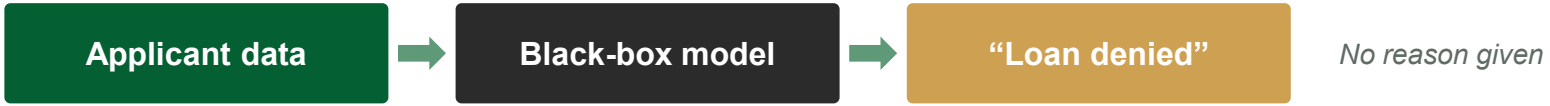
Explainable AI (XAI)

What is explainable AI?

Explainable AI (XAI) is a collection of procedures and techniques that make a machine learning model's outputs understandable and trustworthy for the people who use them.





Interpretable vs explainable: an interpretable model can be read directly (a short rule list, a linear model); an explainable approach adds a method that describes a model we cannot read.

Where it fits: XAI is part of making AI fair, accountable and transparent, especially in health, finance and law.

Benefits of explainable AI

The value of XAI lies in making models transparent enough that humans can understand, check and trust their predictions.

1 Improved decision-making

Shows which factors drive a prediction, so people can combine it with their own knowledge.

Example: A clinician checks why a patient is flagged.

2 Increased trust and acceptance

People adopt models they can question and understand.

Example: Loan officers accept a model that gives reasons.

3 Reduced risks and liabilities

Reveals problems early and supports regulatory and ethical checks.

Example: Spotting a postcode used as a stand-in for ethnicity.

Explanations are one of the best ways to find data leakage and shortcuts the model has learned.

Principles of explainable AI

XAI principles are guidelines for building and deploying models whose behaviour people can see, understand and hold to account.

1 Transparency

We can see which factors influence the model's predictions.

What does the model use?

2 Interpretability

The explanation is clear and usable for its audience.

Can this person understand it?

3 Accountability

Decisions can be checked, questioned and answered for.

Who answers for this decision?

Audience matters: a data scientist, a doctor and a patient need different explanations of the same prediction.

The three components of an XAI system

An explainable AI system combines three components: the machine learning model, an explanation algorithm, and an interface that presents the explanation to people.

1 Machine learning model → **2 Explanation algorithm** → **3 Interface**

Makes the prediction

Example: Predicts a 72% risk that a loan will not be repaid

Works out why

Example: Debt ratio added most to the risk, then short credit history

Shows it to people

Example: Screen message: “Main reasons: debt ratio, credit history”

Don't forget the interface: a correct explanation that people cannot read helps no one.

Global and local explanations

A global explanation describes how the model behaves overall. A local explanation describes why the model made one particular prediction.

Global explanation

“What drives the model overall?”

Asked by: developers, regulators

Example: Glucose and BMI matter most for every patient's predicted risk.

Local explanation

“Why this prediction?”

Asked by: patients, applicants, doctors

Example: This patient's risk is high mainly because of her glucose level.

Interpretable by design

An interpretable (intrinsically explainable) model is simple enough that people can read how it works directly, such as a linear model or a small decision tree.

Feature	Coefficient	Plain meaning
Glucose (per 10 mg/dL)	+0.34	Odds of diabetes × 1.4
BMI (per 5 units)	+0.29	Odds × 1.3
Age (per 10 years)	+0.18	Odds × 1.2
Blood pressure	−0.02	Almost no effect

Logistic regression for diabetes risk: read the model directly (illustrative).

Readable models

- **Linear / logistic regression:** one coefficient per feature.
- **Small decision tree:** a few if-then rules.
- **Rule lists:** ordered if-then rules.
- **Explainable boosting machine:** accurate, still readable.

Try simple first: in high-stakes settings, use a readable model if it is nearly as accurate (Rudin, 2019).

XAI approaches

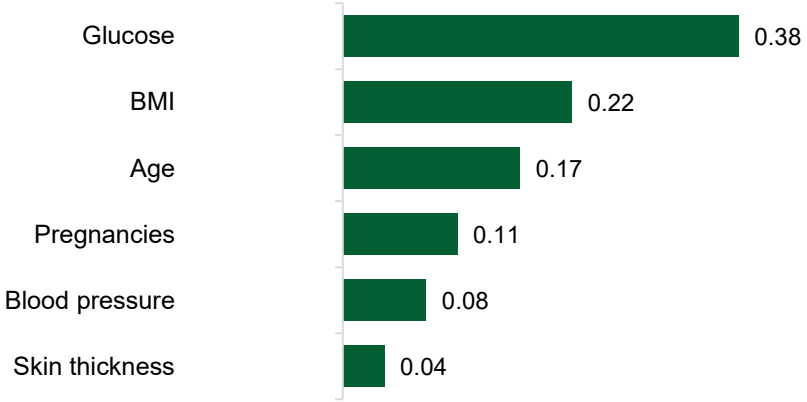
Key question

What kinds of explanation can we give?

Approach 1: feature importance

Feature importance identifies and ranks how much each input feature matters to a model's predictions overall. It is a global explanation.

Feature importance (diabetes model, illustrative)



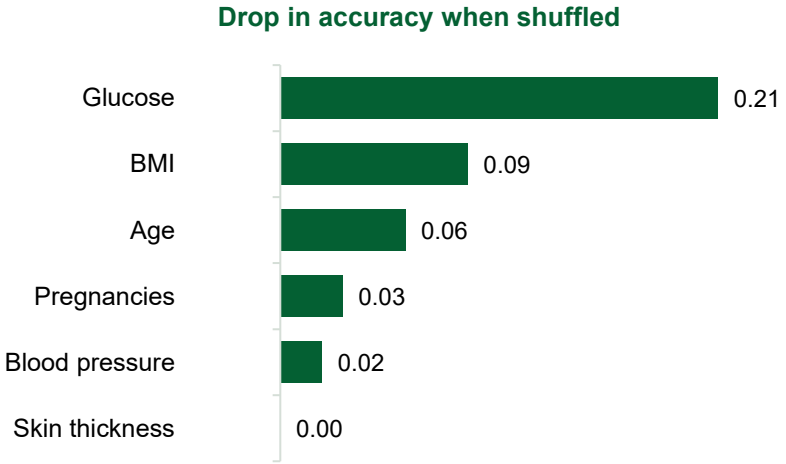
What it tells us

- **Ranking:** which features the model relies on most.
- **Global:** the same ranking for every patient.
- **Not direction:** it says how much, not up or down.
- **Two ways to measure:** permutation (any model) and tree-based Gini importance.

Uses: check the model uses sensible features, spot leakage, and decide which data to collect.

Measuring feature importance: permutation

Permutation importance measures how much a model's score drops on held-out data when one feature's values are shuffled; a large drop means the model relies on that feature.

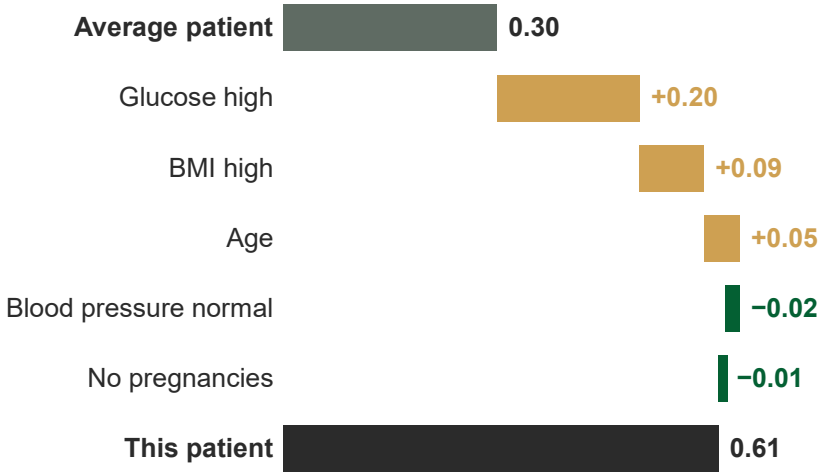


1. **Score:** the model on held-out data.
2. **Shuffle:** one feature's column, breaking its link to the target.
3. **Score again:** with the same model.
4. **Importance:** = the drop. Repeat for every feature.

Cautions: correlated features share importance, and a feature the model relies on is not necessarily a cause.

Approach 2: attribution

Attribution explains one prediction by splitting the gap between that prediction and the average prediction into a contribution from each feature. It is a local explanation.



Everyday analogy: an exam score

- **Class average:** 60.
- **Homework:** +15. **Attendance:** +8.
- **Late submissions:** −3.

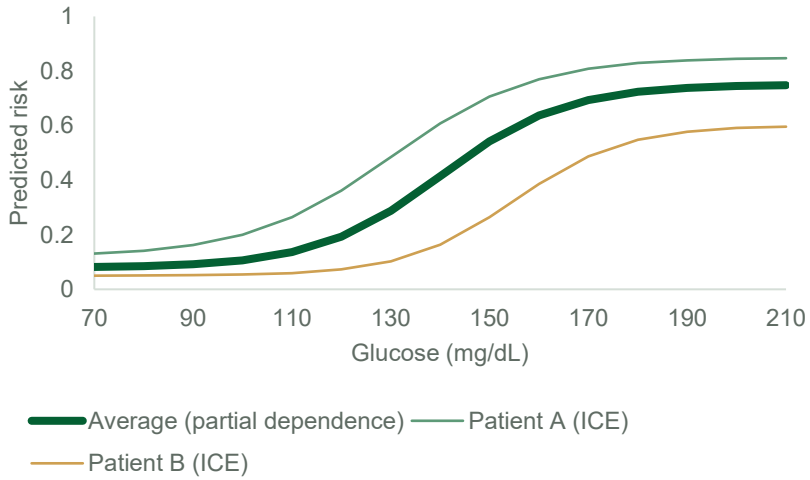
60 + 15 + 8 − 3 = 80

Attribution does the same for a prediction: start from the average and add each feature's push.

Pieces add up: 0.30 + 0.20 + 0.09 + 0.05 − 0.02 − 0.01 = 0.61. Gold pushes risk up, green pulls it down.

Approach 3: visualization

Visualization approaches show a model's behaviour in pictures, such as how the prediction changes as one feature changes, or which parts of an image mattered.



Common visual explanations

- **Partial dependence plot:** average prediction as one feature changes.
- **ICE curves:** the same for each individual; different shapes reveal interactions.
- **Saliency maps:** highlight the image pixels that drove a prediction.
- **Here:** risk rises steeply between about 120 and 160 mg/dL.

Caution: with correlated features, partial dependence averages over unrealistic combinations.

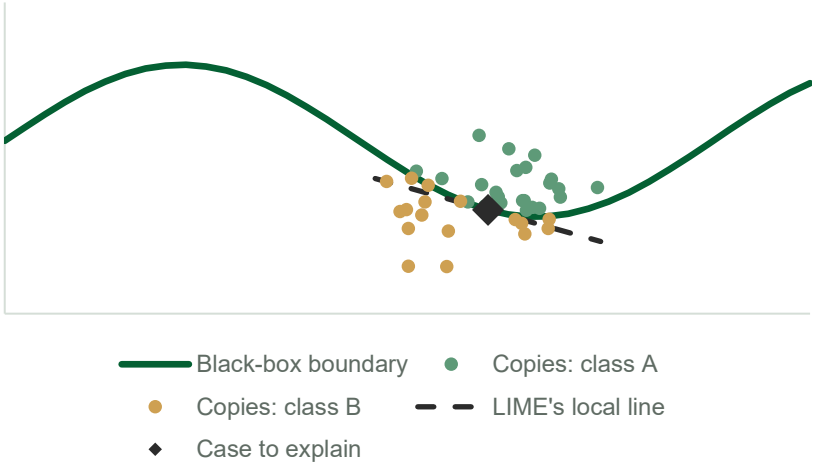
XAI techniques

Key question

Which tools produce explanations, and how?

LIME

LIME (local interpretable model-agnostic explanations) explains one prediction by fitting a simple, usually linear, model that imitates the black box only near that case. It is an approximation, not a replacement.



How it works

- Make many slightly changed copies of one case.
- Ask the black box to predict each copy.
- Weight copies by closeness; fit a simple linear surrogate.
- Read the surrogate's weights as the explanation.

Settings you choose

- **num_features:** how many features to show.
- **num_samples:** how many perturbed copies to create.
- **kernel_width:** how quickly closeness falls off: what “local” means.

Strength: works with any model and with tables, text and images.

Caution: an approximation near one case only; results vary between runs.

Shapley values

A Shapley value is a player's fair share of a team's payout: its extra contribution, averaged over every possible order in which players could join the team.

Order of joining	Ana adds	Ben adds
Ana first, then Ben	$60	$40
Ben first, then Ana	$80	$20
Average = Shapley value	**$70**	**$30**

$70 + $30 = $100 (the whole bonus)

The story

- **Together:** Ana and Ben earn $100.
- **Alone:** Ana would earn $60, Ben $20.
- **Question:** what is a fair split?
- **Answer:** average each person's extra contribution over every order of joining.

Next: replace the friends with features, and the bonus with how far a prediction is from the average.

SHAP: Shapley values for features

SHAP (SHapley Additive exPlanations) computes attribution with Shapley values: features are the players and the prediction is the payout, so each feature gets its fair share.

Order features added	Glucose adds	BMI adds
Glucose, then BMI	0.48 − 0.30 = 0.18	0.56 − 0.48 = 0.08
BMI, then glucose	0.56 − 0.38 = 0.18	0.38 − 0.30 = 0.08
SHAP value (average)	**0.18**	**0.08**

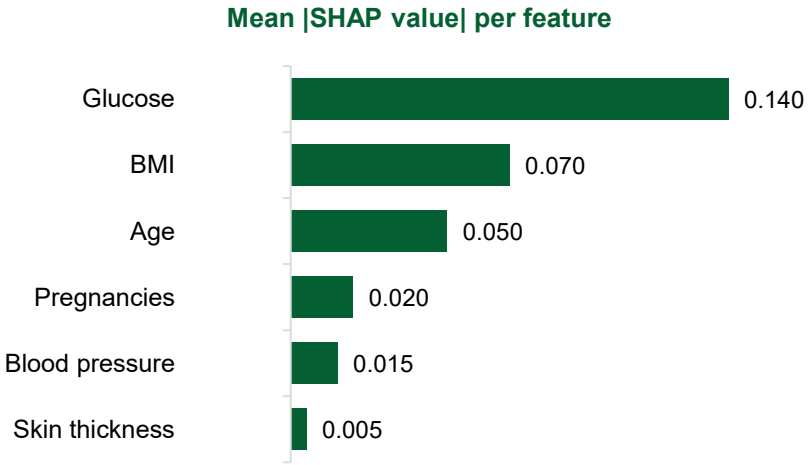
0.30 + 0.18 + 0.08 = 0.56

Game	Model
Players	Features
Payout	Prediction − average prediction
Team without a player	Prediction without that feature known
Fair share	SHAP value

Adds up to the model's output: for classifiers this is often log-odds rather than probability, so check before reading the numbers.

From local to global with SHAP

Global SHAP importance is the average size of each feature's SHAP values over many predictions, turning many local explanations into one overall ranking.



Local to global

- **Mean |SHAP|:** average size of each feature's contributions = global importance.
- **Summary plot:** one dot per case; shows size and direction of each effect.
- **Dependence plot:** SHAP value against feature value, like a PDP.
- **Cross-check:** compare with permutation importance.

Agreement builds confidence: if SHAP and permutation importance disagree, look for correlated features.

Limits, cases and the future

Key question

Where does XAI fall short, and where is it going?

In this part

- Limitations
- Case studies
- Future trends

Current limitations of XAI

XAI limitations are the reasons an explanation may be costly, narrow, inconsistent or misleading, even when the method is applied correctly.

1 Computational cost

Exact SHAP or many LIME runs are slow on big models and data.

2 Limited scope

Many methods suit only some models, data types or domains.

3 No standards

No agreed way to measure or compare explanation quality.

4 Methods disagree

LIME and SHAP can rank features differently for one case.

5 Model, not world

A feature that matters to the model need not cause the outcome.

6 Can mislead

A plausible explanation does not prove the model is right.

Top row: the three limitations in GeeksforGeeks. Bottom row: three more that matter in practice.

XAI case studies

XAI case studies show how explanations are used in practice to support diagnoses, text classification and image recognition.

1 Medical imaging

Heat map over an X-ray shows the region behind a “pneumonia” prediction.

Lesson: check it looks at the lungs, not hospital markers.

2 Natural language processing

Highlighted words show what drove a sentiment prediction.

Lesson: check it reads meaning, not shortcuts.

3 Computer vision

Saliency maps show which pixels made the model say “wolf”.

Lesson: it was looking at snow, not the animal.

Common thread: explanations caught models that were accurate for the wrong reasons, which accuracy alone would not reveal.

Future developments and trends

Future XAI developments are the new methods, wider adoption and new rules expected to shape how models are explained.

1 New methods

Explanations in terms of human concepts; methods for large language models.

2 Increased demand

Explanations become a routine part of building and deploying models.

3 Regulation and ethics

The EU AI Act (2024) requires transparency and human oversight for high-risk AI.

Origins: Judea Pearl's work on causality, and LIME (2016), which popularised model-agnostic explanations.

Direction: explanations are becoming a legal as well as a technical requirement.